STUDENT WORKSHEET

Oxygen and sulfur

Compare electrons, shells, and periodic-table positions

Name: _____ Date: _____

Element	Atomic number Z	Ground-state electron configuration
Oxygen (O)	8	[He] $2s^2$ $2p^4$
Sulfur (S)	16	[Ne] $3s^2$ $3p^4$

Use neutral atoms in their ground state. Atomic number **Z** counts protons; a neutral atom has the same number of electrons as protons. **[He]** means $1s^2$ (2 electrons); **[Ne]** means $1s^2$ $2s^2$ $2p^6$ (10 electrons). A leading number, such as 3, identifies the shell *n*; a superscript counts electrons in a subshell. For these atoms, the period number equals the number of occupied shells.

Model reminder: Simplified Bohr-style rings help count electrons by shell. They aren't literal electron paths and don't show orbital shapes or electron probability distributions.

1. Particle counts

Complete the counts for each neutral atom.

	Protons	Electrons
Oxygen		
Sulfur		

2. Read the configurations

Show the electron total for each atom, including its noble-gas core.

O: _____

S: _____

3. Count by shell

Write each atom's electron counts from its innermost shell outward.

O: _____

S: _____

4. Find the periods

Give each element's period. Explain how occupied shells support your answer.

5. Compare the outer shells

Both elements are in group 16. How many outer-shell electrons does each have? Which subshells supply them?

6. Correct a claim

A classmate says, "Sulfur has eight more outer-shell electrons because its Z is eight larger." Correct the claim.

7. Check a shell sketch

A sulfur sketch labels its shells 2, 6, 8, from inside outward. Is it correct for the neutral ground-state atom? Correct it and explain.

8. Use a model carefully

Name one feature a shell diagram helps you compare and one feature it cannot show accurately.

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TEACHER DIRECTIONS AND ANSWER KEY

Oxygen and sulfur

Compare electrons, shells, and periodic-table positions

Suggested use: Introductory chemistry, grades 9-12 (editorial suggestion).

Give students page 1. They need only a pencil and the printed data, with no website or account. Ask them to show electron totals before comparing shells. Review why an atom's total electron count differs from its outer-shell count.

Scope: These questions describe neutral atoms in their ground state. Shell counts are a simplified representation; use the model reminder to discuss what the rings leave out.

1. **Particle counts.** Oxygen: 8 protons, 8 electrons. Sulfur: 16 protons, 16 electrons. Equal positive and negative charges make each atom neutral.
2. **Configurations.** Oxygen: 2 + 2 + 4 = 8. Sulfur: 10 + 2 + 4 = 16. The noble-gas core must be included in the total.
3. **Shell counts.** Oxygen: 2, 6. Sulfur: 2, 8, 6. Expanded sulfur configuration: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$.
4. **Periods.** Oxygen is in period 2, with two occupied shells; sulfur is in period 3, with three occupied shells.
5. **Outer shells.** Each has 6 outer-shell electrons. Oxygen has $2s^2 + 2p^4$; sulfur has $3s^2 + 3p^4$. Both are in group 16.
6. **Claim correction.** Sulfur has 8 more electrons in total, while both outer shells contain 6. Sulfur has an additional occupied shell; its inner-shell counts are 2 and 8.
7. **Sketch correction.** Use 2, 8, 6. The proposed 2, 6, 8 totals 16 but places the electrons in the wrong shells for ground-state sulfur. A correct total alone is not enough.
8. **Model limits.** Accept comparisons of shell counts, occupied shells, or outer-shell counts. For a limit, accept that rings are not literal paths, or that the sketch does not show orbital shapes or electron probability distributions. Accept equivalent accurate wording.

Reference data

Royal Society of Chemistry: Oxygen (Z 8) - https://periodic-table.rsc.org/element/8/oxygen

Royal Society of Chemistry: Sulfur (Z 16) - https://periodic-table.rsc.org/element/16/sulfur

Element facts checked November 1, 2026. Shell counts are derived from the configurations above.